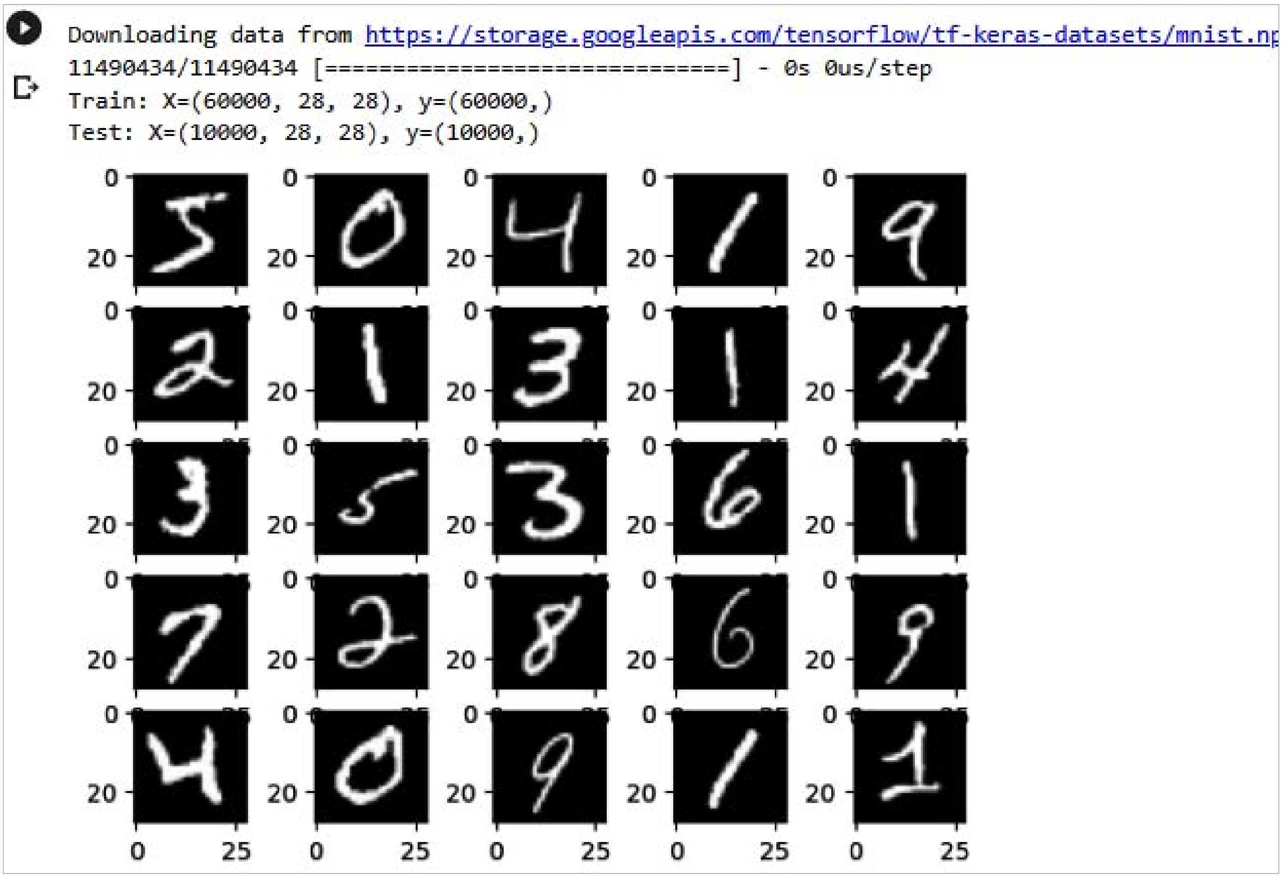

Figure 3: Output of CNN model implemented using MNIST data set

Implementing a CNN deep learning model using TensorFlow

A CNN deep neural network is frequently utilised for image categorisation, object detection, and other computer vision applications. Figure 2 shows the Python code example executed on Google Collab notebook for implementing a CNN deep learning model using TensorFlow.

We have trained the CNN model using the MNIST data set in this example. In the machine learning community, the MNIST data set is frequently used as a benchmark for image classification tasks, notably for assessing the performance of deep learning models like CNNs. It is a popular data set due to its simplicity and accessibility, and has been used for various applications including handwriting recognition, digit classification, and optical character recognition.

The output of our implemented CNN deep learning model using the MNIST data set is shown in Figure 3. The data set consists of 70,000 images, each of which is a grayscale image of a handwritten digit from 0 to 9. The images are 28x28 pixels and centred within a 32x32-pixel image.

TensorFlow is a Google open source software library. It was designed for tasks requiring complex numerical computations, and is popular because it supports Python and C++ API. It has faster compile times and supports CPUs as well as GPU distributed processing. Overall, TensorFlow's flexibility, scalability, and facilitation of neural network architectures make it well-suited for various industry applications. END

References

- *https://www.tensorflow.org/api_docs*
- *https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz*
- *https://www.tensorflow.org/tutorials/images/cnn*

By: Dr Aditya Bhardwaj

The author is a B. Tech, M. Tech, and PhD in CSE. He works as assistant professor in the School of Computer Science Engineering and Technology at Bennett University, Greater Noida. He is experienced in cloud computing and open source technology.